

Ethics Statement

This study is a retrospective secondary analysis of de-identified clinical imaging (CT volumes and intra-operative bronchoscopy videos). The clinical data were de-identified by the responsible hospital department through its strict institutional review process and were provided by Dr. Nan Wei. Data access and use were reviewed and authorised through the institutional data-governance process of Henan Provincial People’s Hospital. No animal experiments, prospective enrolment, research-directed intervention, or identifiable participant information were involved, and the study complies with the Declaration of Helsinki.

Public benchmarks (BOP Linemod/T-LESS, SCARED, real-colon FEM, EndoNeRF, Blended-MVS) are used under their original providers’ access terms; these contain no new human or animal material acquired for this study. The clinical CT-to-bronchoscopy evaluation uses de-identified patient CT and intra-operative video provided by Henan Provincial People’s Hospital under the authorisation described above.